Latent Feature Extraction for Musical Genres from Raw Audio

GTZAN Dataset Evaluation and Improvements

UE24CS352A Machine Learning - Mini-Project Write-up | Team: Harsha K - PES1UG24CS666, Sunil - PES1UG24CS634
Code: github.com/Harsha-Kamaraj/ML-Mini-Project
Final notebook: **musical_genre_classification.ipynb**

1 Problem statement

Classify short music clips into **10 genres** and learn a compact 64-dimensional genre embedding. We implement the models of Sawhney, Vasavada and Wang [1], extend the experiment to the full GTZAN dataset, and evaluate four existing improvement stages. Raw waveforms are used through Step 3; Step 4 adds mel spectrograms. The earlier four-genre architectures and training settings are retained, with classifier outputs expanded from 4 to 10. The paper's four-class scores are historical context, not directly comparable ten-class benchmarks.

2 Dataset and preprocessing

We use **all 1,000 GTZAN recordings, 100 per genre** [2]: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae and rock. All recordings decode successfully; they are approximately 30 seconds long at 22,050 Hz. A pinned mirror archive is verified by SHA-256, extracted and converted from .au to .wav. Twenty-eight recordings share identical decoded audio with another recording.

For the paper models, each song supplies 20 consecutive segments of 20,000 samples (0.907 s). Average pooling over 40 samples gives 500-dimensional inputs, without normalisation: **20,000 clips total**. Improved models standardise inputs using training data only. Every song supports 33 segments; when selected by the dev sweep, the extra 13 segments are added only to training. On the song split, these come only from training songs.

3 Approach and evaluation

Paper models

Two-layer NN: 500 -> 128 (tanh) -> 10, trained with cross-entropy. **Vanilla AE:** 500-256-192-128-64-128-192-256-500, trained with squared-L2 reconstruction loss. **Deep Softmax AE:** the same autoencoder with a 64-32-16-10 classifier and $L = gamma\ ||x - reconstruction||^2 + (1 - gamma)\ CE$, gamma = 0.9. Adam uses learning rate 0.0001, batch size 512 and dropout keep probability 0.9. Epoch limits are 200, 400 and 500, respectively. Settings follow the paper, using the linked code where the paper is silent. The vanilla AE is assessed by reconstruction, not classification accuracy.

Improvement stages

Step 1 - fixed AE: linear decoder output, train-only standardisation, epoch reshuffling, best-dev checkpointing, early stopping and mean-squared reconstruction loss; the dev sweep chooses gamma = 0.1. **Step 2 - more signal:** the sweep retains pool size 40 and adds the remaining 13 segments per training song under the song split.

Step 3 - convolutional raw audio: four strided 1-D convolution blocks with 16, 32, 64 and 128 channels, global average/max pooling, a 64-dimensional encoding, transposed-convolution decoder and 10-class head. The selected pool size is 4. **Step 4 - mel spectrograms:** 2-D convolution blocks on 64 x 40 log-mel inputs (FFT 1,024; hop 512). Both use gamma = 0.1, Adam at 0.001, batch size 64, at most 40 epochs and patience 8.

Evaluation protocol

Clip split: 15,000/2,500/2,500 train/dev/test clips before extra training segments. Segments of a song can cross partitions; seed 0 has 935 test songs also present in training. **Song split:** targets 800/100/100 songs (80/10/10 per genre; exact at seed 0). Identical audio is grouped globally, including a metal/rock duplicate, so test songs and exact recording duplicates are unseen during training. Each dev/test song contributes 20 clips.

Six classifiers are evaluated with seeds 0-4 on both splits: **60 selected-model evaluations**. Sweeps choose hyperparameters on the clip-split dev set at seed 0; these settings carry over to both protocols. The displayed best model uses mean clip-split dev accuracy. We report test clip accuracy, macro-F1 and song accuracy (mean softmax probabilities over a song's test clips).

4 Implementation

The final notebook contains acquisition, all models, dev sweeps, tables, confusion matrices, PCA of 64-dimensional encodings and 10-dimensional logits, and an audio demo. All **33 code cells** completed. Completed CPU experiments were retained; 18 remaining experiments ran on a Colab **Tesla T4** (Python 3.13.15, PyTorch 2.11.0+cu130, LibROSA 0.11.0). The notebook falls back to CPU locally. With **REUSE_COMPLETED = True** and **RETRAIN = False**, saved results are reused from the two ten-genre artifact folders. Missing entries start training.

5 Results on all 10 genres

Table 1 reports **mean ± sample standard deviation over five seeds**. Both middle columns evaluate individual test clips; the last column aggregates predictions per held-out song. All rows are our ten-class implementations.

Model	Clip split clip accuracy	Song split clip accuracy	Song split song accuracy
Paper two-layer NN	17.1% ± 0.8%	17.2% ± 2.1%	21.8% ± 4.7%
Paper Deep Softmax AE	28.0% ± 1.0%	27.3% ± 1.9%	39.6% ± 4.6%
Step 1: fixed Deep Softmax AE	27.6% ± 0.3%	26.6% ± 1.6%	36.0% ± 4.7%
Step 2: more signal	30.2% ± 0.7%	29.2% ± 1.9%	37.2% ± 8.2%
Step 3: convolutional raw audio	70.6% ± 1.8%	61.2% ± 2.0%	74.8% ± 5.1%
Step 4: mel spectrograms	**87.3% ± 0.8%**	**73.4% ± 5.7%**	**82.6% ± 6.3%**

Table 1. Source: artifacts/improvements/gtzan_10_genres/all_results.csv (60 rows); the same values are summarised in comparison.csv. The vanilla AE has no classification head.

Paper-model findings. The Deep Softmax AE reaches 28.0% clip-split test accuracy and 27.3% on the song split; the two-layer NN reaches about 17% on both. In Part 1's seed-0 joint model, the fraction of positive reconstruction outputs is 0.0. This observed decoder collapse motivates the linear-output correction. Step 1 alone does not improve mean test accuracy; Step 2 raises it to 30.2% / 29.2%.

Improved-model findings. The convolutional raw-audio model reaches **70.6% / 61.2%** clip accuracy on clip/song splits. The mel model is best at **87.3% / 73.4%**, and is selected by mean clip-split dev accuracy (87.8%). On unseen songs, its macro-F1 is **73.1% ± 5.6%**; aggregating clip probabilities gives **82.6% ± 6.3% song accuracy**, compared with **74.8% ± 5.1%** for raw audio.

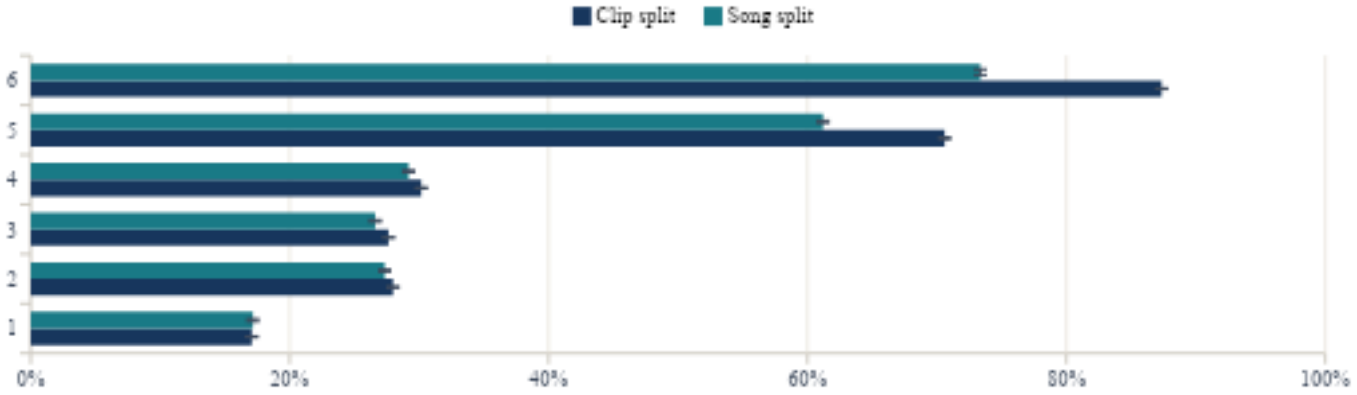


Figure 1. Test clip accuracy for the same six classifiers. Error bars show standard deviation across five seeds. The chart contains only the full ten-genre experiment.

6 Conclusions and limitations

Representation matters. The convolutional raw-audio pipeline exceeds Step 2 by 32.0 percentage points on the song split; mel spectrograms add another 12.2 points. The mel model is the strongest tested classifier, while the raw model retains the waveform-only approach. These are comparisons of complete pipelines, not controlled estimates of each change's effect.

Evaluate unseen songs. Clip-split accuracy exceeds song-split accuracy by **9.4 points for raw audio and 14.0 points for mel spectrograms**. Shared-song clips make the clip split easier; song-disjoint results are the more useful estimate for new recordings. Gamma = 0.1 wins the single-seed dev sweeps, but this does not establish a general benefit from reconstruction or isolate its contribution.

Limits and reproducibility. GTZAN has only 1,000 recordings, with about 100 held-out songs per seed. Exact duplicates are grouped, but artist overlap and other dataset faults [3] are not fully controlled, and no external dataset was tested. Reported deviations measure variation over seeds/splits, not confidence intervals. The final run combines completed CPU and GPU experiments; recorded scores are preserved, while new predictions or training can change with package versions and hardware. Preserve the artifacts and recorded environment for reproduction.

References. [1] A. Sawhney, V. Vasavada, W. Wang. *Latent Feature Extraction for Musical Genres from Raw Audio.* Stanford University, 2018. [2] G. Tzanetakis, P. Cook. *Musical Genre Classification of Audio Signals.* IEEE Trans. Speech and Audio Processing, 2002. [3] B. L. Sturm. *The GTZAN dataset: Its contents, its faults, their effects on evaluation, and its future use.* arXiv:1306.1461, 2013.